

America Gaona Borges

828-702-9909 • gaonaborgesa@berea.edu • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

EDUCATION

Berea College | *Berea, KY*

Expected May 2026

B.A in Computer & Information Science • GPA: 3.7/4.0

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, Python, HTML/CSS

Frameworks & Libraries: React, NestJS, Node.js, LangChain, Prisma, AWS S3

AI / ML: Retrieval-Augmented Generation (RAG), pgvector, Ollama, OpenCV, Neural Networks

Tools & Platforms: Git, GitHub, Docker, Figma, PostgreSQL, Swagger UI, JTAG, SEGGER Embedded Studio

Embedded Systems: Microcontroller Programming, Firmware Development, SPI, TWI, UART, BLE

EXPERIENCE

Software Engineer Intern | NextGen Justice - *San Jose, CA*

Jun 2025 – Aug 2025

- Developed a full-stack, AI-powered legal assistance platform using NestJS, PostgreSQL, React, and Docker, architecting RESTful APIs following a Model-Controller-Service (MCS) pattern with Swagger UI documentation.
- Identified gaps in AI response quality and built a dedicated prompt template service with LangChain to enhance a Retrieval-Augmented Generation (RAG) pipeline, improving context-aware AI responses for legal document reasoning.
- Led end-to-end product design: defined user flows and product requirements, translated them into hand-sketched wireframes and Figma prototypes, before frontend implementation in React.
- Created a TypeScript endpoint to generate and store document embeddings using Ollama and pgvector, enabling scalable semantic search across legal documents.

Lead CS Undergrad Teaching Assistant | Berea College - *Berea, KY*

Aug 2024 – Present

- Mentored 25 new TAs through onboarding and weekly coordination; led tutoring sessions for 20+ students in robotics and game design, contributing to a 30% grade improvement vs. the previous year – managing cross-functional stakeholders (faculty, TAs, students) to align on outcomes.
- Contributed to the creation of the CS department website using HTML, CSS, and JavaScript, creating a centralized hub for students exploring the CS major.

Robotics Engineering Intern | Purdue University - *West Lafayette, IN*

May 2024 – Jul 2024

- Built a smart container using embedded systems and ML algorithms to analyze eating habits; programmed nRF52 microcontrollers in Embedded C, integrating gyro, accelerometer, and temperature sensors.
- Debugged firmware using JTAG and SEGGER Embedded Studio, improving system performance and reliability.
- Authored and presented a paper on communication protocols to 20+ peers.

PROJECTS

ML Autonomous Navigation System | Python, OpenCV, C++, BFS, SLAM

Built an autonomous robot navigation system combining computer vision, a custom-trained neural network for handwritten digit recognition, BFS path planning, and SLAM localization. Achieved 3cm positioning accuracy.

AI-Powered Legal Platform (NextGen Justice) | NestJS, React, LangChain, pgvector, Figma

Full-stack application with RAG pipeline, semantic search via document embeddings, and a fully designed UI from wireframes to Figma to production React components.

UI/UX Product Research | Figma, User Research, Usability Testing, Participatory Design

End-to-end product research project with a human-centered focus to improve how CS students discover career opportunities. Conducted participatory design sessions, usability testing, and delivered optimized interface prototypes in Figma.

SHPE-Berea Official Website | HTML/CSS, JavaScript, Bootstrap | Live at: shpebera.github.io

Designed and built the official responsive SHPE-Berea website with team profiles, event carousels, and membership information.

LEADERSHIP & COMMUNITY

President | Society of Hispanic Professional Engineers

Nov 2024 – Present

Public Relations Officer | Girls Who Code

Aug 2023 – Present

Language Program Volunteer | Hispanic Outreach Program

Aug 2023 – Dec 2023